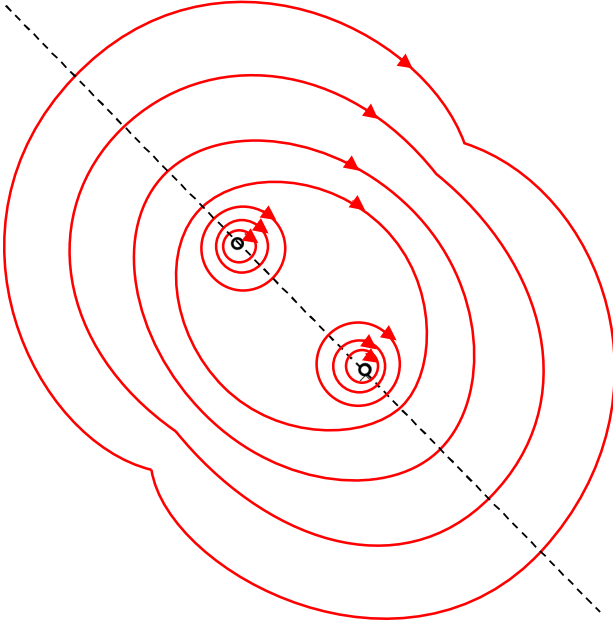
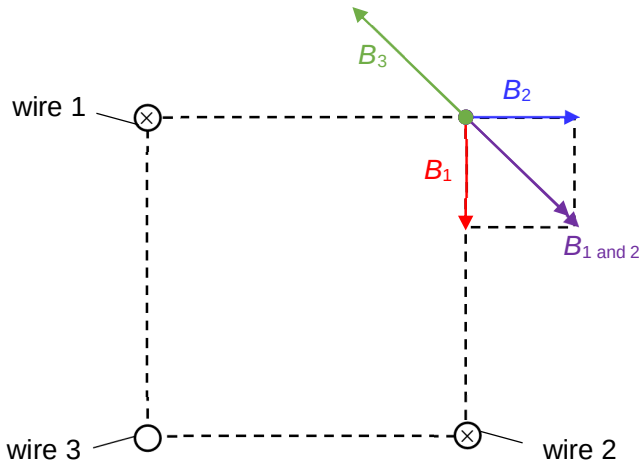


	highly energetic so may damage onboard equipment.	A0
5(a)	force per unit length per unit current acting on a straight conductor carrying the current placed normal to the magnetic field.	B2
5(b)(i)	 shape - nearly-concentric circles then symmetric ovals/figure-of-8s clockwise arrows check for increasing distance along dotted line	B1

Qns	Answer	Marks
5(b)(ii)	direction: out of paper	B1
	magnitude of B due to wires 1 and 2 individually:	C1
	$B = \frac{\mu_0 I}{2\pi d}$ $= \frac{\mu_0 I_s}{2\pi L}$	
	vector sum of B due to 1 and 2 is 45 degree to south of east:	C1
	$B_{\text{due to 1 and 2}}^2 = \left(\frac{\mu_0 I_s}{2\pi L}\right)^2 + \left(\frac{\mu_0 I_s}{2\pi L}\right)^2$ $B_{\text{due to 1 and 2}} = \sqrt{2} \left(\frac{\mu_0 I_s}{2\pi L}\right)$	
	at empty corner, magnitude of magnetic flux density due to wire 3 = vector sum of magnetic flux density due to wire 1 and wire 2	M1
	$B_{\text{due to 3}} = \sqrt{2} \left(\frac{\mu_0 I_s}{2\pi L}\right) = \frac{\mu_0 I_3}{2\pi \sqrt{2} L^2}$	
	$\left(\frac{I_s}{I_3}\right) = \frac{1}{2} = 0.5$	A1 (do not accept fractions)
		

Qns	Answer	Marks
6(a)	Faraday's Law: Induced emf is directly proportional to the rate of	B1